Change Detection

IRS Chapter 15

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Quiz references: **Change Detection: Interpreting Bitemporal and Multitemporal Evidence** (Week 10).

Introduction to Change Detection in Remote Sensing

- **Change detection**: Identifies changes in surface features over time.
- Key applications:
  - Land use change
  - Vegetation monitoring
  - Environmental management

Bitemporal vs. Multitemporal Change Detection

- **Bitemporal Detection**:
  - Compares images from two specific dates.
  - Suitable for detecting abrupt changes.
- **Multitemporal Detection**:
  - Analyzes multiple dates for trend analysis.
  - Ideal for gradual or seasonal changes.
- Increasingly accessible data enables broader applications.

Conditions for Effective Change Detection

1. Compatible sensors, acquisition times, pixel sizes, and viewing angles.
2. Similar seasons and vegetation growth stages.
3. Accurate co-registration, preferably within 0.2 pixels (Dai and Khorram, 1999).
4. No clouds or cloud shadows in the analysis area.
5. Convert to top-of-atmosphere (TOA), preferably bottom-of-atmosphere (BOA / surface), reflectance. Atmospheric scattering and absorption can mimic surface change; TOA alone does not remove them.
6. Exclude irrelevant conditions: soil moisture differences when studying canopy cover, or different harvest dates when studying long-term vegetation change.

Preprocessing Steps for Change Detection

This is the preparation checklist before the visual and numerical comparisons that follow.

- **Calibrate**: Convert sensor measurements to compatible radiometric units.
- **Co-register**: Align corresponding ground locations on a common pixel grid.
- **Correct reflectance**: TOA normalizes illumination geometry; BOA additionally estimates atmospheric effects to recover surface reflectance.
- **Mask clouds and shadows**: Exclude obscured pixels rather than inventing surface observations.
- **Control other variation**: Match season and moisture conditions; normalization must not erase the change being studied.

Why Atmospheric Correction Matters

Scattering adds light from outside the target; absorption changes the measured spectrum.

TOA reflectance still contains atmospheric effects. BOA / surface reflectance corrects estimated atmospheric contributions, with residual uncertainty.

Use the same processing level on both dates. Haze differences can mimic NDVI change without land-cover change.

Source: USGS / Michelle Bouchard, “Landsat 8 Surface Reflectance example,” May 3, 2013. Public domain.



Visual Interpretation in Change Detection

Start with the paired images before choosing a numerical method. The next examples compare the same ground locations at two dates.

- **Multidate Color Composite**: Combines bands from two dates.
- **Image Swiping/Flickering**: Overlay images for change visualization.
- **Side-by-Side Comparison**: View images in parallel for analysis.
- **Heads-Up Digitizing**: Manually identify and classify changes.

Image Swiping/Flickering Example

Lombardia, Italy: Sentinel-2 patch 1208, selected dates in 2019



Gallo et al., Sentinel-2 Image Time Series for Crop Mapping (CC0); shared RGB stretch

Image Swiping/Flickering Example

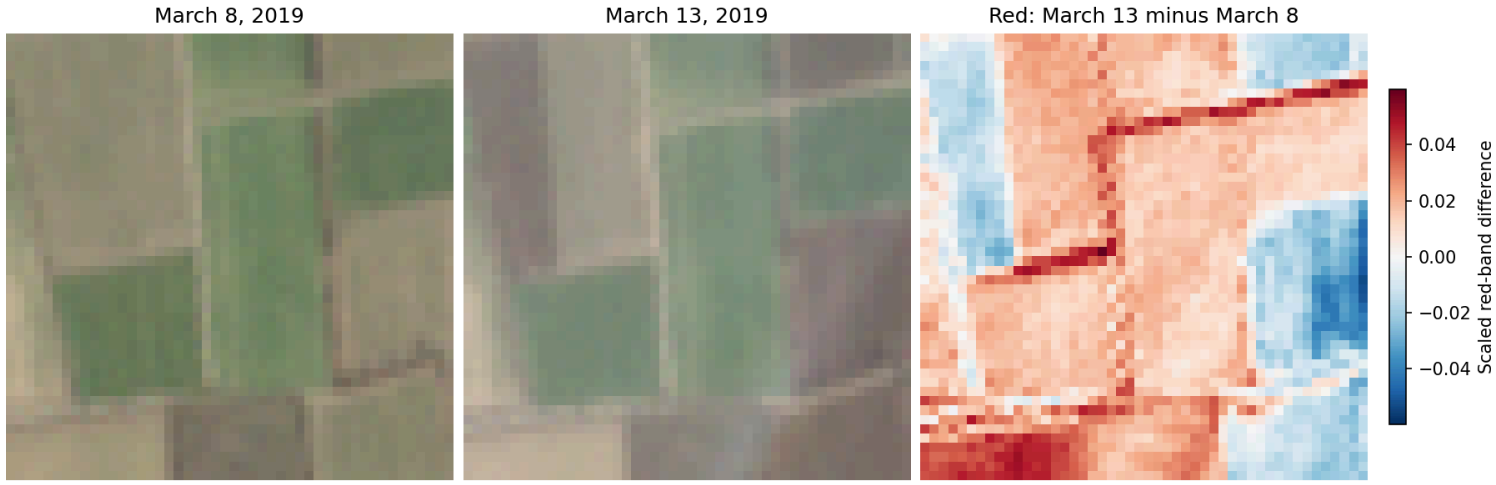


Follow the same field boundaries through the sequence. Every frame uses the same RGB stretch.

Open the animation

Comparing Two Dates: Lombardia

$$\Delta\rho_{red} = \rho_{red,\text{March 13}} - \rho_{red,\text{March 8}}$$



- **Side-by-side**: use the same RGB stretch on both dates. **Difference**: blue indicates red-band decrease; red indicates increase, with a neutral midpoint.
- Put **zero at 50% of the colormap**, not 25%, using symmetric limits $[-M, M]$. Equal positive/negative magnitudes should look equally strong. Give masked pixels a separate appearance, not the zero color.

Run the Lombardia notebook

Sentinel-2, Lombardia3 patch 1208; March 8 and 13, 2019. Stored values scaled by 0.0001.

Image Algebra for Change Detection

Now compare **time slices** at corresponding ground locations. The inputs can be a band, a derived index, or a spectral vector from each date; computing NDVI within one date is a separate step.

- **Image Differencing**: Subtract corresponding values between dates.
- **Image Ratioing**: Divide corresponding values across dates; denominators near zero are unstable.
- **Euclidean Distance**: Measure the size of the multiband difference.

$$D = \|\mathbf{x}_1 - \mathbf{x}_2\|_2 = \sqrt{\sum_{b=1}^{B} (x_{1,b} - x_{2,b})^2}$$

Here $\mathbf{x}_1$ and $\mathbf{x}_2$ contain the same bands at one location on dates 1 and 2. Repeat the calculation at each valid location to produce a change raster.

Linear Algebra Notation and Inner Product

Red-band differencing compares one number per pixel. For CVA, each pixel becomes a **vector of band values**. We need vector lengths and angles to describe multispectral change.

Bold symbols such as $\mathbf{x}$ denote vectors. The **inner product** (dot product or scalar product) is:

$$\mathbf{x}^T\mathbf{y} = \mathbf{x}\cdot\mathbf{y} = \sum_{i=1}^{n} x_i y_i$$

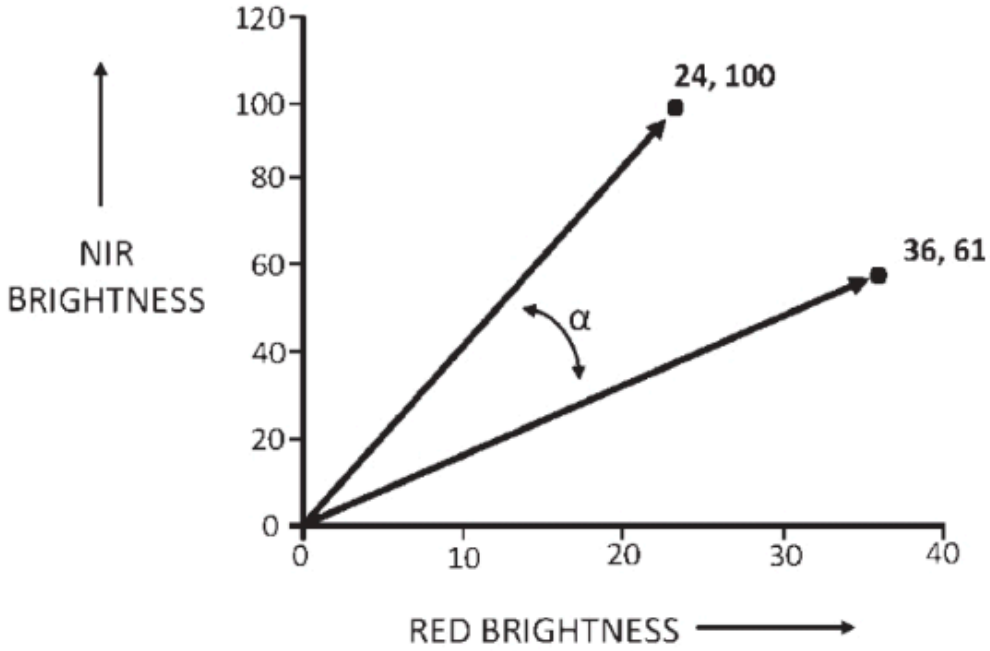
$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^T\mathbf{x}}, \qquad \cos\theta = \frac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}$$

For nonzero vectors, θ is their angle: 0° means aligned; 90° means perpendicular.

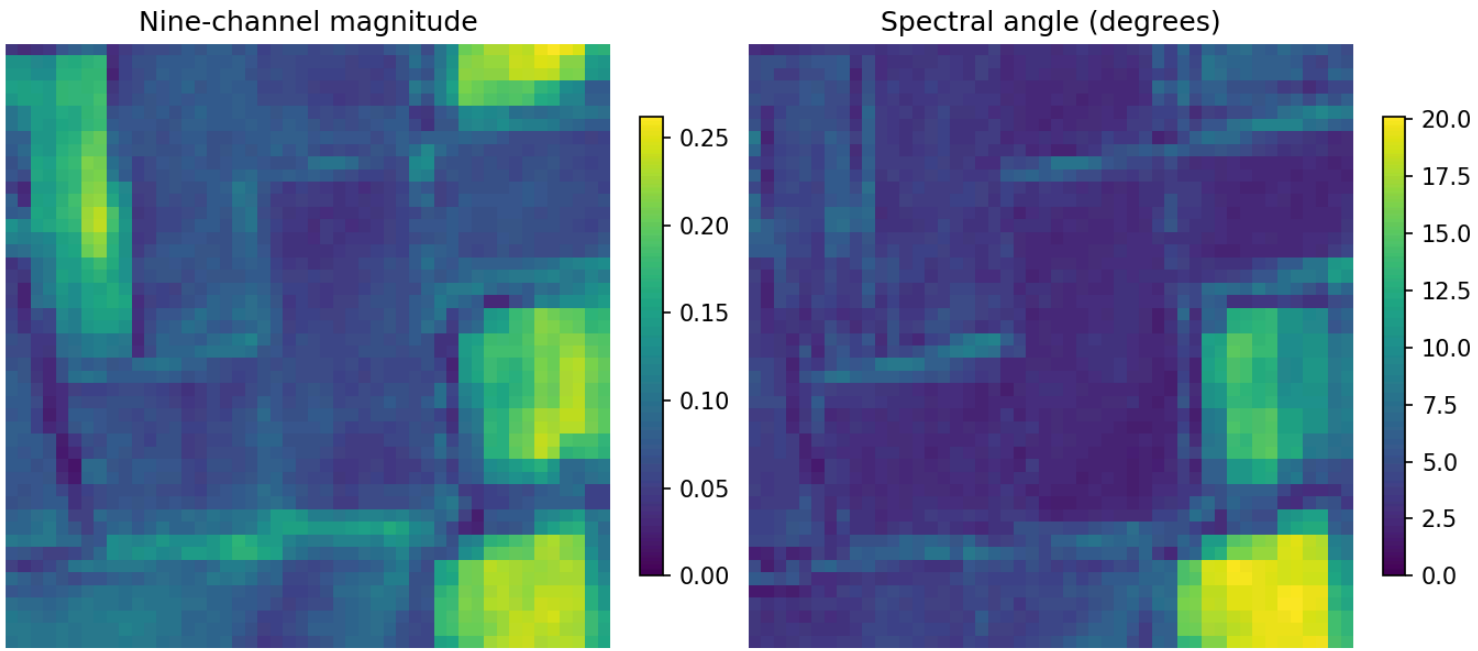
Change Vector Analysis (CVA)

- **Magnitude**: Size of the multispectral change vector.
- **Direction**: Which bands increased or decreased; useful for interpreting change.
- Spectral angle is a separate similarity measure, not CVA's change-vector magnitude.

$$\Delta \mathbf{x} = \mathbf{x}_{t_2} - \mathbf{x}_{t_1}, \qquad \|\Delta \mathbf{x}\|_2 = \sqrt{\sum_{b=1}^{B} (x_{t_2,b} - x_{t_1,b})^2}$$



Change Vector Analysis (CVA)



- **Magnitude (left):** choose when the absolute size of the band-value change matters. Detects overall brightening/darkening as well as spectral-shape changes; requires comparable radiometry and band scales.
- **Spectral angle (right):** choose when relative band proportions matter more than overall brightness. Multiplying all bands by the same positive factor leaves the angle unchanged, so it can miss real brightness-only change.

Transformation and Data Reduction

So far, we compared band values directly. To focus on vegetation-related change, transform both dates into the same features, such as greenness or wetness, **then compare across dates**.

Review: Chapter 11, PCA and Tasseled Cap (Equation 11.9).

$$\begin{bmatrix} B \\ G \\ W \end{bmatrix} = \mathbf{C}_{\text{sensor,product}}\boldsymbol{\rho}, \qquad B = \sum_b c_{B,b}\rho_b, \quad G = \sum_b c_{G,b}\rho_b, \quad W = \sum_b c_{W,b}\rho_b$$

Use the **same coefficients and band order** on both dates, matched to the sensor and reflectance product. Brightness, greenness, and wetness are fixed axes; PCA axes depend on the fit.

$$\Delta G = G_{t_2} - G_{t_1}, \qquad \mathbf{z} = \mathbf{P}^T(\boldsymbol{\rho} - \boldsymbol{\mu})$$

PCA Review: One Shared Coordinate System

Previously covered: pixels are samples, bands are features. Fit one rotation to both dates; independently fitted PC1 axes need not represent the same direction.

```
from sklearn.decomposition import PCA
import numpy as np

# Each array contains valid pixels by bands, in the same band order.
model = PCA(n_components=3)
model.fit(np.vstack([pixels_march8, pixels_march13]))
pc_march8 = model.transform(pixels_march8)
pc_march13 = model.transform(pixels_march13)
pc_difference = pc_march13 - pc_march8
```

PCA maximizes variance, not change relevance. Inspect what each component represents before interpreting its difference.

Classification in Change Detection

The output is a **spatial label map**, with a class assigned to each valid pixel or segment.

- **Postclassification comparison**: classify each date separately, producing two land-cover maps. Compare labels to produce transitions such as forest-to-forest or forest-to-pasture.
- **Direct change classification**: concatenate both dates' bands and train on transition labels, producing one transition map directly.
- **Targeted attribution**: first detect changed locations; classify the transition only there. Retain unchanged and invalid locations as separate states.

Choosing the labels is part of defining the task. Binary change and named transitions answer different questions.

What Classification Produces

The book emphasizes **attributing change**, not just detecting it: did lost forest become pasture, a strip mine, or suburban development?

- **Classify the whole area**: combine bands from both dates into one feature vector per pixel or multitemporal segment. Assign a change class where the two-date signatures are separable.
- **Classify only detected changes**: first locate changed areas, then assign a class describing the change there.

The spatial result labels locations by the change of interest, such as **forest-to-pasture**, rather than reporting only a spectral distance. Which labels can be distinguished depends on the data and classification rule.

Statistical Techniques: Image Cross-Correlation

Within a forest patch, some pixels are brighter than others. After disturbance, that spatial pattern may change. **Image cross-correlation measures how strongly the two dates' pixel values vary together.**

1. Choose one band and a neighborhood or multitemporal segment.
2. Pair each valid pixel's date-1 value with its date-2 value at the same ground location.
3. Plot date 1 horizontally and date 2 vertically: each location contributes one point.
4. Calculate Pearson correlation r for those points. Repeat across segments to supply spatial evidence for change detection or classification.

A tight upward-sloping scatter indicates strong positive correlation; a dispersed scatter indicates weaker linear association.

Calculating Image Cross-Correlation

Let $x_{t,p}$ be one band's value at date t and paired location p, for n valid locations in a segment. $\bar{x}_t$ is its spatial mean.

$$s_t = \sqrt{\frac{\sum_{p=1}^{n}(x_{t,p} - \bar{x}_t)^2}{n-1}}, \qquad \text{cov}_{12} = \frac{\sum_{p=1}^{n}(x_{1,p} - \bar{x}_1)(x_{2,p} - \bar{x}_2)}{n-1}$$

$$r = \frac{\text{cov}_{12}}{s_1 s_2} = \frac{1}{n-1}\sum_{p=1}^{n}\left(\frac{x_{1,p} - \bar{x}_1}{s_1}\right)\left(\frac{x_{2,p} - \bar{x}_2}{s_2}\right)$$

Subtracting the mean expresses each pixel's relative brightness within its segment. Dividing by SD puts both dates on a common scale. Positive products come from locations that are above both means or below both means; these increase r.

One s_1, s_2, and r per segment. Correlation is undefined if either SD is zero.

Using Correlation, Slope, and Intercept

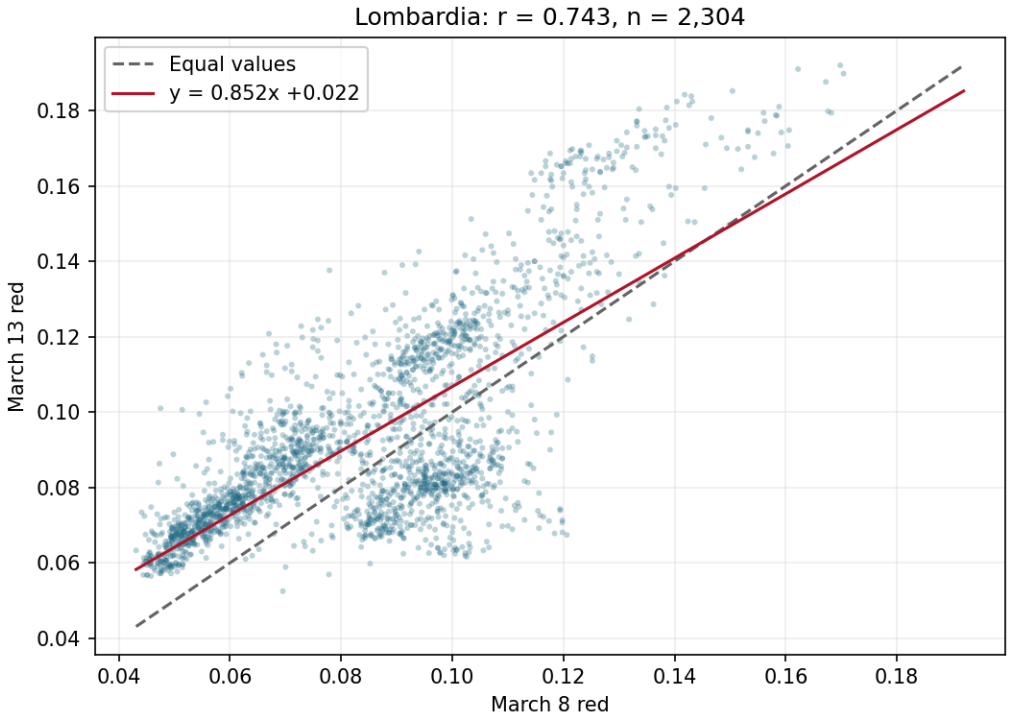
The book combines descriptions of the **relationship between dates** as evidence for change:

- **Correlation** r: strength and sign of the linear association. Values near $+1$ indicate a consistent bright/dark pattern; values near 0 indicate weak linear association; values near -1 indicate a reversed pattern.
- **Regression slope**: how strongly differences among locations on date 1 carry through to date 2.
- **Regression intercept**: the fitted brightness offset between dates.

Example: if every date-2 value equals $2x_1 + 0.1$, then $r = 1$, slope $= 2$, and intercept $= 0.1$. Together they describe a preserved pattern with changed contrast and brightness.

Use these features in classification or further statistical analysis. Accurate registration and cloud masking help ensure that the paired values describe the same surfaces.

Example: Cross-Correlation of Lombardia Red Values



Each point pairs the same location on March 8 and 13. The fitted line is $y = 0.852x + 0.022$; $r = 0.743$ summarizes the positive linear association across this patch.

Run the paired-pixel calculation in Colab

One region: Lombardia3 patch 1208. Inspect the paired RGB views alongside these statistics.

Multitemporal Change Detection

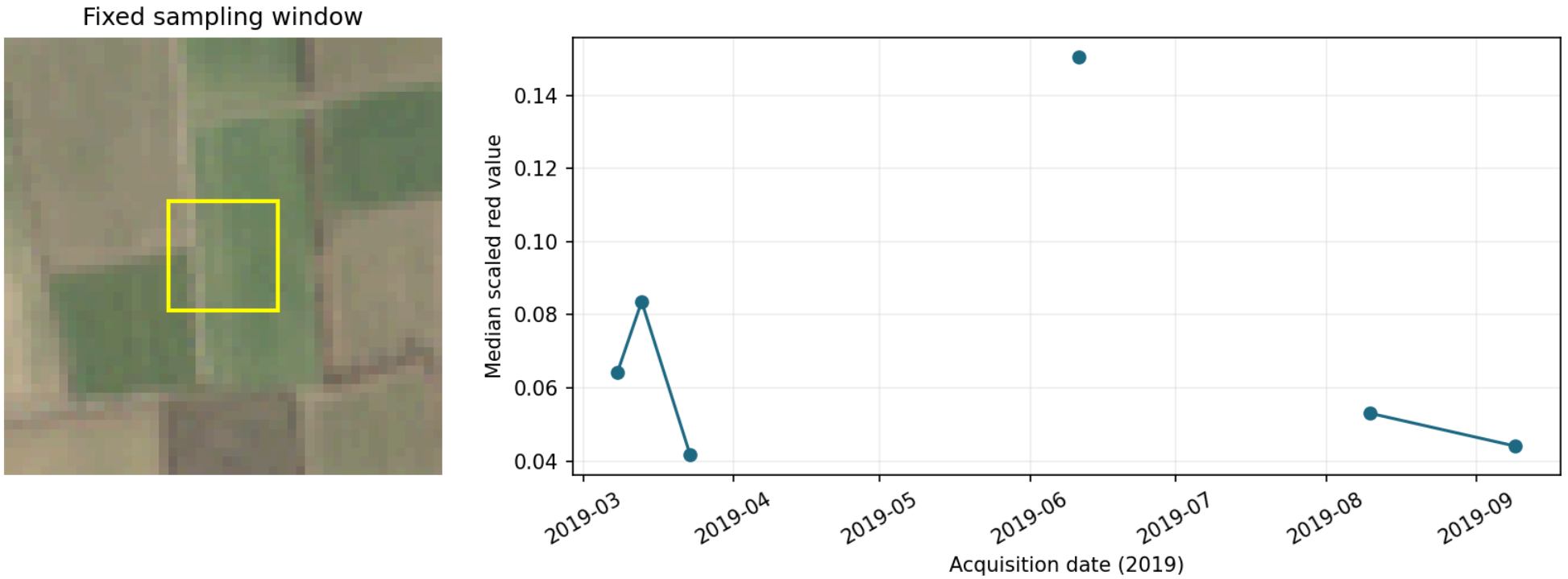
Benefits

- **Enhanced Change Detection**: Detects low-magnitude trends.
- **Gradual Change Capture**: Monitors slow processes (e.g., vegetation growth).
- **Aligned with Socioeconomic Cycles**: Annual analysis supports policy planning.

Challenges

- **Cloud Cover**: Persistent clouds complicate analysis.
- **Data Volume**: Large datasets require robust processing.
- **Phenological Effects**: Seasonal changes can obscure results.

Following a Fixed Location Through Time



The marked window stays fixed. Plot its median red-band value against acquisition date to connect the image sequence to a numerical trajectory.

Use the RGB sequence to interpret vegetation, soil, and atmospheric contributions to these observations.

Lombardias patch 1203, selected 2019 dates; same sampling window and scaling throughout

Coming Next: Preparing a Time Series

Roadmap: first handle missing observations and unwanted variation, then fit seasonal behavior and monitor departures.

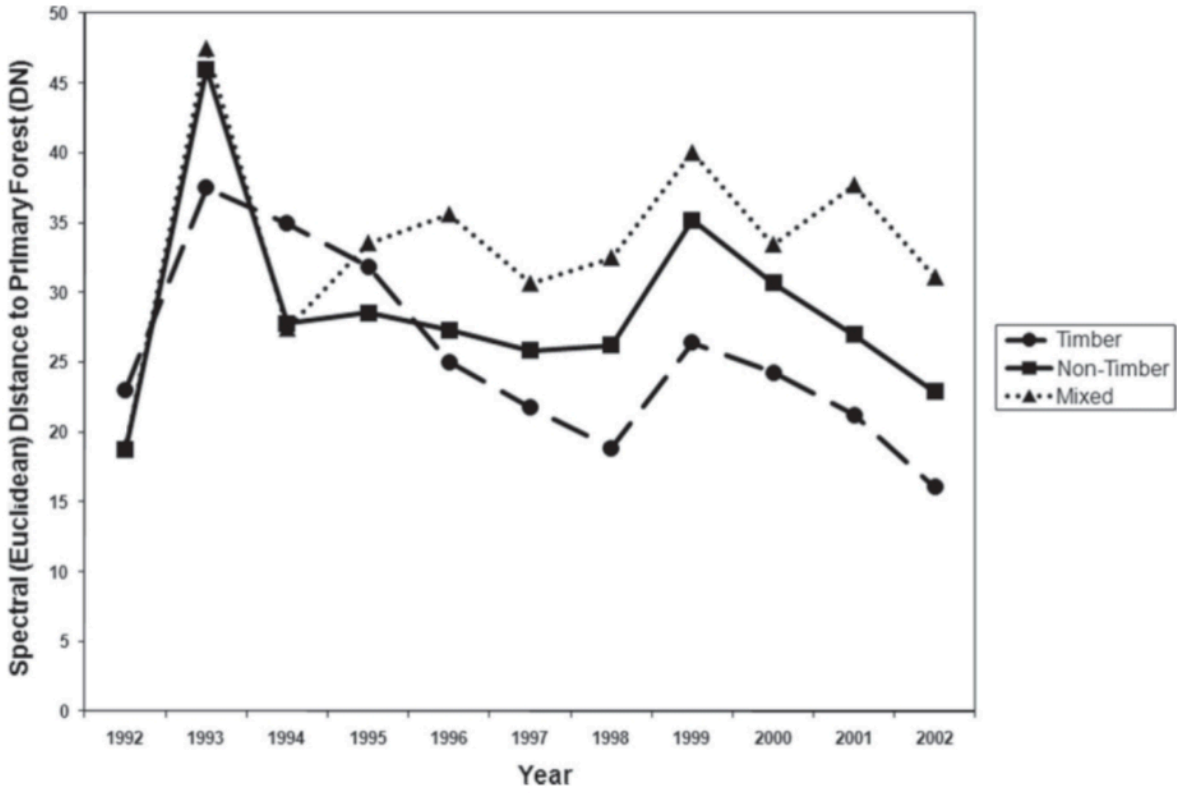
- **Take Care Collecting Images**: Aim for the same season and time of day.
- **Image Co-Registration**: Align ground locations; histogram matching is radiometric normalization, not registration.
- **Cloud Removal**: Manual and automated techniques.
- **Image Normalization**: Adjusts for atmospheric or seasonal variance.
- **Segmentation (optional)**: Groups pixels with similar trajectories.
- **Band Selection (optional)**: Focus on a single band or derived value to simplify analysis.

Cloud Cover Analysis in Multitemporal Change Detection

- **Challenge**: Cloud cover limits clear imagery, especially with one image per year.
- **Image Selection**: Choose the clearest annual image available.
- **Gap Handling Options**: (To fill the hole left when clouds are removed)
  - **Pixel Interpolation**: Use nearby pixels in the same image.
  - **Date Interpolation**: Use nearby dates' images for missing areas.
  - Keep missing observations explicit; interpolation can hide abrupt changes.

EWMA = exponentially weighted moving average. It averages new observations with a running history, giving recent observations more weight. This helps detect small, persistent departures that a noisy single observation could hide. It monitors available observations; it does not fill cloud gaps. We derive it later in the control-chart section.

Example from the book....



Mean spectral distance from primary forest by agroforestry group

Time Series Analysis Basics

Roadmap for this section: we will work through basis-function regression, then seasonal harmonics and residual monitoring. Top-down and bottom-up methods are alternatives listed for context, not derived here.

- **Kernel Regression**: Linear combination of basis functions.
- **Top-Down Methods**: Whole-series approximation, then partition.
- **Bottom-Up Methods**: Builds larger structures from smaller ones.

Least Squares (LS) Solution for Basis Functions

- **Goal**: Minimize error between observed values $ y $ and model predictions $\hat{y}$.
- **Model Representation**: Express $\hat{y}_x$ as a sum of basis functions:

$$\hat{y}_x = a_0 f_0(x) + a_1 f_1(x) + \cdots + a_n f_n(x)$$

- **Objective**: Find coefficients a_i that minimize the sum of squared residuals.

Example: Monomial Basis Functions in LS Solution

- **Monomial Basis Functions**: Use powers of x: $1, x, x^2, x^3$.
  - $f_i(x) = x^i$
- **Observations:** $\{(x_i, y_i) \mid \quad i = 1 \ldots m\}$
  - Note I use x rather thant t, whatever!
- **Three Cases**:
  - **Underdetermined**: Fewer points than functions (insufficient data).
  - **Exactly Determined**: Same number of points as functions.
  - **Overdetermined**: More points than functions (more data than needed).

Setting Up the Matrix for LS Solution

- **Matrix Setup**: Each element (i, j) is $f_j(x_i)$.

$$\mathbf{A} = \begin{bmatrix} f_0(x_1) & f_1(x_1) & \dots & f_n(x_1) \\ f_0(x_2) & f_1(x_2) & \dots & f_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_0(x_m) & f_1(x_m) & \dots & f_n(x_m) \end{bmatrix}$$

- **Objective**: Solve $\mathbf{Aa} = \mathbf{y}$ for coefficients $\mathbf{a}$.

Solving with the Normal Equations

- **Normal Equations**: Used when the system is overdetermined.

$$\mathbf{A}^T\mathbf{A}\mathbf{a} = \mathbf{A}^T\mathbf{y}$$

- **Interpretation**: Projects $\mathbf{y}$ onto the space spanned by the basis functions.
- **Underdetermined Systems**: Require additional constraints or regularization.

$$(\mathbf{A}^T\mathbf{A} + \lambda\mathbf{I})\mathbf{a} = \mathbf{A}^T\mathbf{y}$$

Harmonic Basis Functions

- **Cosine and Sine Functions** as Basis:

$$\hat{\rho}_x = a_0 + a_1 \cos\left(\frac{2\pi}{T}x\right) + a_2 \sin\left(\frac{2\pi}{T}x\right)$$

- **Periodicity**: Fits seasonal changes in time series data.
- **Coefficients** a_0, a_1, a_2: Control the overall value and seasonal oscillation.
- T **(Period)**: Defines the cycle length, e.g., $T = 365$ for annual data.

Solving for Harmonic Coefficients

- **Objective**: Minimize the sum of squared residuals between observed data y_x and model predictions $\hat{\rho}_x$.
- **Set Up**: Rewrite model as a matrix equation.

$$\begin{bmatrix} 1 & \cos\left(\frac{2\pi}{T}x_1\right) & \sin\left(\frac{2\pi}{T}x_1\right) \\ 1 & \cos\left(\frac{2\pi}{T}x_2\right) & \sin\left(\frac{2\pi}{T}x_2\right) \\ \vdots & \vdots & \vdots \\ 1 & \cos\left(\frac{2\pi}{T}x_n\right) & \sin\left(\frac{2\pi}{T}x_n\right) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

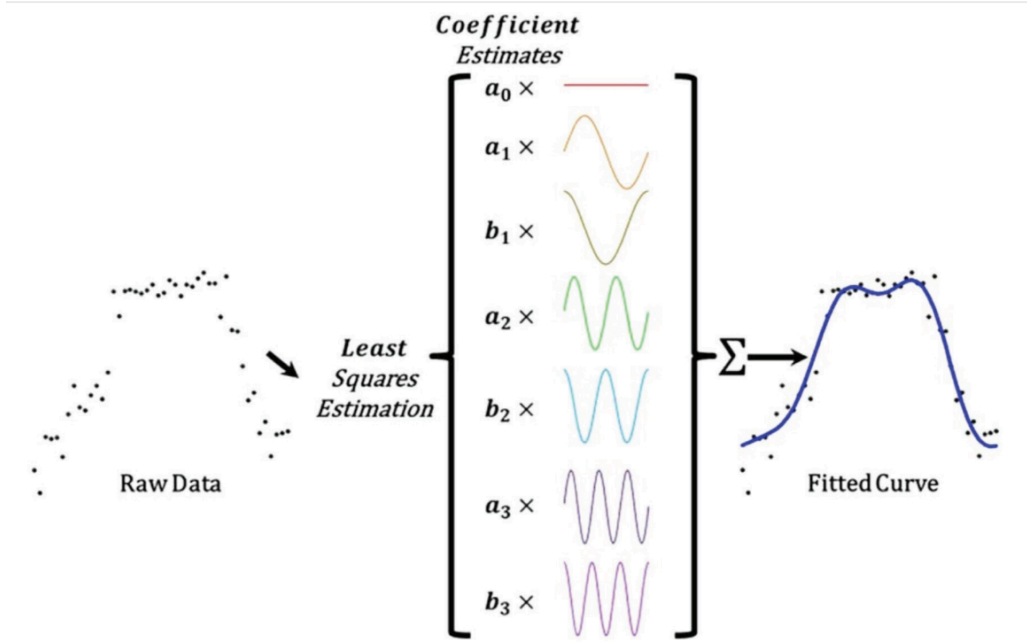
- **Solution**: Use normal equations: $\mathbf{A}^T\mathbf{A}\mathbf{a} = \mathbf{A}^T\mathbf{y}$.

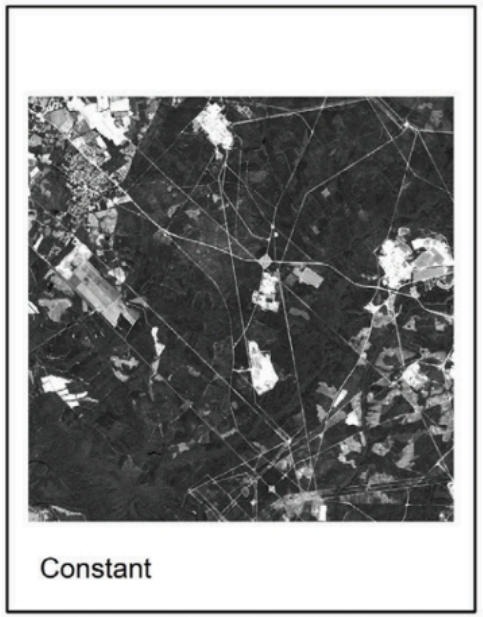
Harmonic Regression for Change Detection

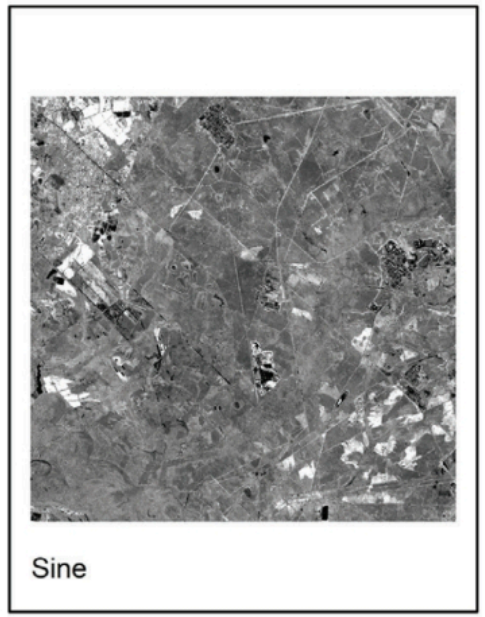
- **Models Seasonal Trends:**

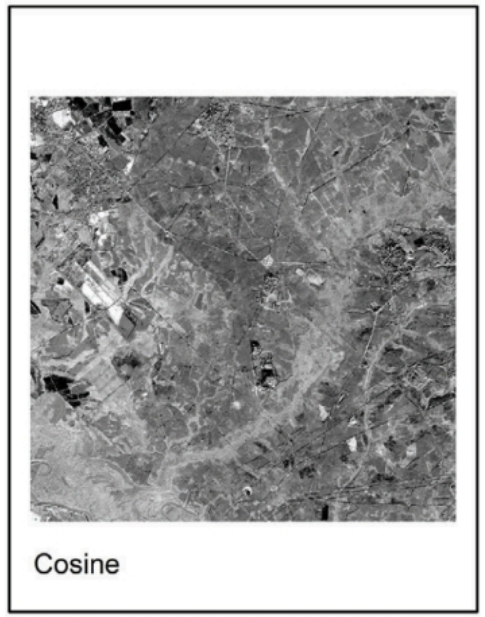
$$\hat{\rho}_x = a_0 + a_1 \cos\left(\frac{2\pi}{T}x\right) + a_2 \sin\left(\frac{2\pi}{T}x\right)$$

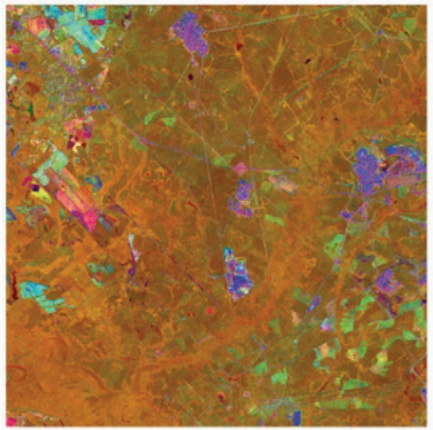
- Note that you can add new bases for different periods (months, seasons, weeks, etc)



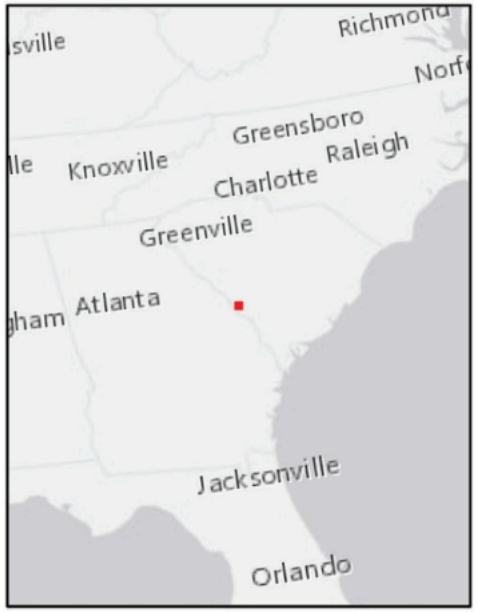

Constant


Sine


Cosine


RGB (Cosine, Sine, Constant)





Harmonic Basis and Fourier Transform

- **Fourier Series**: Generalizes harmonic regression by capturing multiple frequencies.

$$\hat{\rho}_x = a_0 + \sum_{k=1}^{N} \left(a_k \cos\left(\frac{2\pi k}{T} x\right) + b_k \sin\left(\frac{2\pi k}{T} x\right) \right)$$

- **Fourier Transform**: Decomposes signals into sine and cosine functions.
- **Fast Fourier Transform (FFT)** computes the discrete Fourier transform efficiently in $O(N \log N)$; it does not solve general least-squares regression coefficients.
- **Relation to Harmonic Regression**:
  - Harmonic regression approximates with a single frequency.
  - Fourier transform reveals all significant frequencies in data.

Why Average the Residuals?

A slightly low NDVI value could be measurement noise. Several low values in succession are stronger evidence of a persistent shift.

1. Fit the expected seasonal cycle using a stable reference period.
2. Calculate residuals $e_t = y_t - \hat{y}_t$: observed minus expected NDVI.
3. Average recent residuals to detect a sustained departure without treating the seasonal cycle itself as disturbance.

An **exponentially weighted moving average (EWMA)** keeps this running history with a single number.

EWMA: Recent Evidence Gets More Weight

$$z_t = \lambda e_t + (1-\lambda) z_{t-1}, \qquad 0 < \lambda \leq 1$$

e_t is the newest residual; z_t is the smoothed residual. Start at the baseline mean, $z_0 = 0$.

With $\lambda = 0.2$, each update uses 20% new evidence and 80% previous history. Residuals receive weights $0.2, 0.16, 0.128, \ldots$ as they age: the weights decay geometrically.

If residuals remain -0.10, the first three averages are -0.020, -0.036, and -0.0488. Repeated departures accumulate evidence.

Smaller λ smooths more and responds more slowly. At $\lambda = 1$, there is no smoothing.

Coming Next: Monitoring With Control Charts

Coming next: smooth the seasonal-model residuals with EWMA, set control limits, and interpret the forest-thinning example.

- **Purpose**: Monitor changes in time series data, such as vegetation or land cover.
- **Types of Control Charts**:
  - **EWMA (Exponentially Weighted Moving Average)**: Detects small, gradual shifts.
  - **Shewhart X-bar Chart**: Detects large, abrupt shifts.

The next few slides go into more detail.

EWMA Control Charts with Tolerance Bands

- Monitor the smoothed residual z_t rather than the raw seasonal measurement.

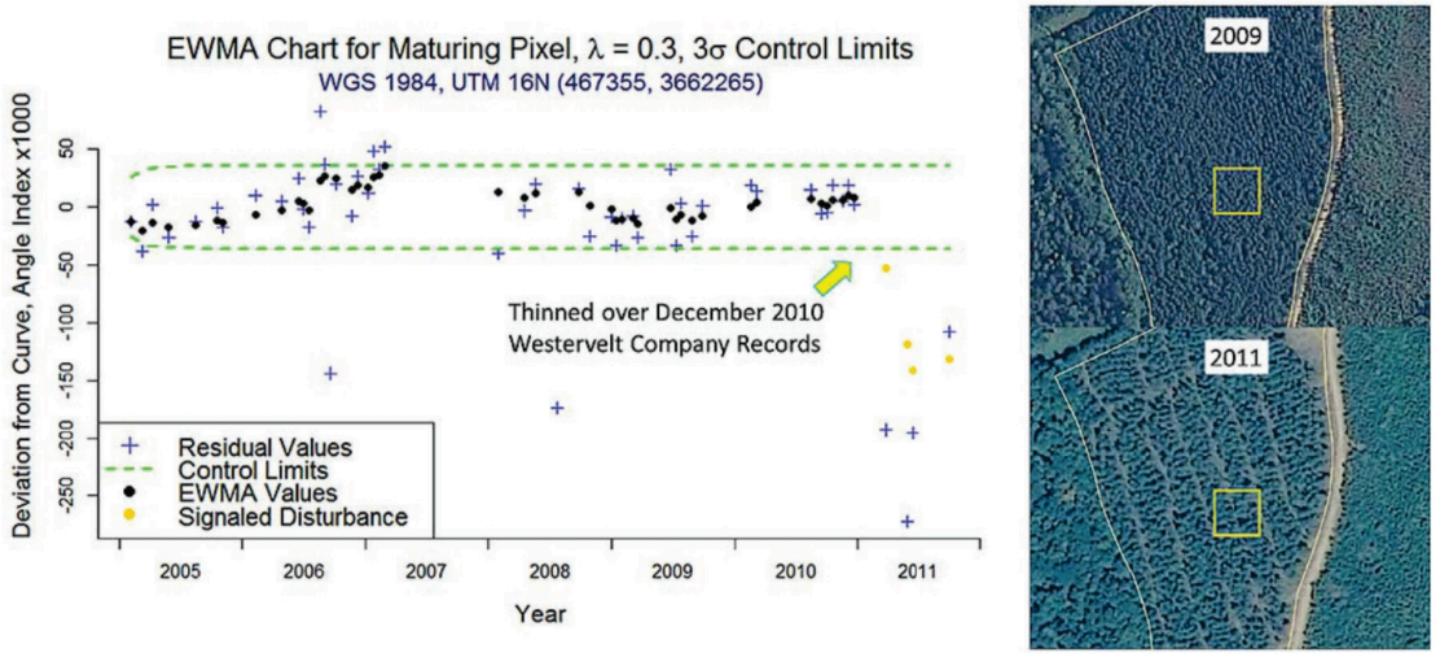
$$z_t = \lambda e_t + (1 - \lambda) z_{t-1}$$

- First fit seasonality, calculate residuals, and monitor those residuals.
- **Control Limits (Tolerance Bands)**:
  - Define acceptable variation range.
  - Persistent departures can signal small or gradual shifts as well as abrupt disturbances.

Applications

- **Anomaly Detection**: Highlights abrupt shifts, such as unexpected environmental changes.
- **Preprocessing for Trend Analysis**: Filters noise to stabilize data for further analysis.

Figure 15.7: EWMA Detects Forest Thinning



Blue crosses: seasonal-model residuals. Black points: EWMA, $\lambda = 0.3$. Green dashed lines: 3σ control limits. Yellow points: signaled disturbance.

After thinning in **December 2010**, repeated negative residuals pull the EWMA below the lower limit. The yellow boxes locate the same pixel in the 2009 and 2011 NAIP images.

Vertical axis: deviation from the fitted curve, angle index multiplied by 1000, not raw NDVI. Source: Brooks et al. (2014), textbook Figure 15.7.

Notebook: Compare Two Change Maps

Run the Lombardia demo in Colab

View the notebook as slides

Compare the signed red-band difference, nine-channel magnitude, and spectral angle. Relate the maps to the RGB pair, then follow the same patch through the year's image sequence.

Summary of Change Detection in Remote Sensing

- Change detection supports environmental and urban planning.
- **Bitemporal**: Suitable for abrupt changes.
- **Multitemporal**: Captures long-term trends.
- Increasing data availability enhances capability.